

Code in Place 2025

Stanford CS106A

Section - Week 4

Python Control Flow

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Today's Agenda



1. Check-In

How are we all doing?



2. Concepts Review

Control Flow



3. Practice Problem

"High Low Game"

Please Turn On Your Camera



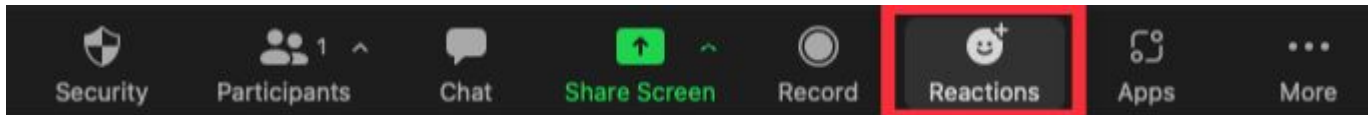
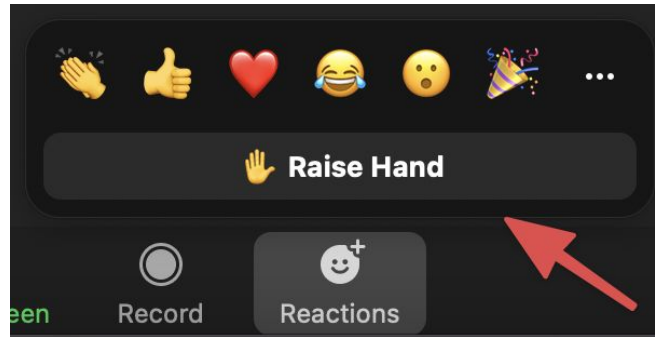
If you're able, please **turn on your camera!**
.... It can really make the section come to life!



(Image source: <https://as.virginia.edu/eight-ways-get-more-out-zoom>)

Zoom Reactions

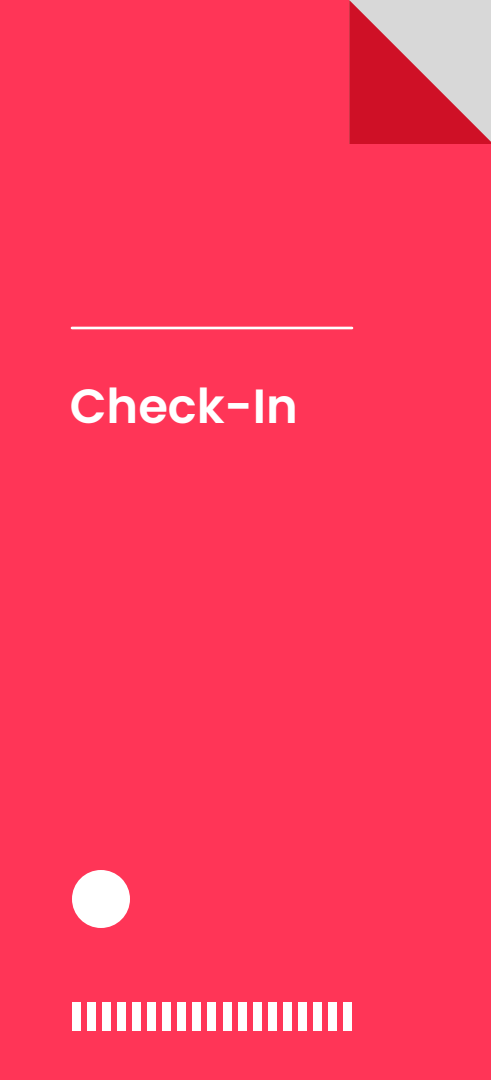
- 👍 **Thumbs Up:** If you understand.
- 🙋 **Raise Hand:** If you have a question (or just speak in the mic).



Introductions

Hi, I'm **David!**

- Head TA, here at my 4th Code in Place.
 - Started as a CIP student! 2x volunteer Section Leader.
- CS @ Massachusetts Institute of Technology (MIT)
- Product Manager and former Software Engineer
- Love photography, video games, movies
- Guilty pleasure: Reality competition shows like Survivor



Check-In

How is everyone doing?
Hopefully your third week of CIP went well!

- Share something nice that happened in the last week

Concepts Review

Control Flow



- Everything from Karel is still in play!
- for loops
- while loops
- booleans (True or False, used in conditional statements)
- if, else, elif

```
user_number = int(input("Enter a number: "))    # Variable assignment
if user_number == 0:                            # Equality comparison
    print("Your number is 0!")
elif user_number > 0:
    print("Your number is positive!")
else:
    print("Your number is negative!")
```


Expressions & Arithmetic Operators



- Any **expression** on the right side of the equal sign is calculated before being assigned to a variable.

- Precedence of operations** (similar to “PEMDAS” in Algebra):
 - `()` “parentheses” **highest precedence**
 - `**` “exponentiation”
 - `-` “negation”
 - `*`, `/`, `//`, `%`
 - `+`, `-` **lowest precedence**



- Example:

```
x = (1 + 3 * 5 / 2) * (-3)    # Python will calculate the right-side expression.
x = (1 + 15 / 2) * (-3)
x = (1 + 7.5) * (-3)
x = (8.5) * (-3)
x = -25.5                    # float -25.5 assigned to “x”.
```

Comparison Operators



Operator	Meaning	Example	Result
==	equals	1 + 1 == 2	True
!=	does not equal	3.2 != 2.5	True
<	less than	10 < 5	False
>	greater than	10 > 5	True
<=	less than or equal to	126 <= 100	False
>=	greater than or equal to	5.0 >= 5.0	True



WARNING: Notice the difference between variable **assignment** vs **equality** comparison.

- `x = 64` # Assigns the int value 64 to the variable x.
- `x == 64` # Checks if the value of x is equal to 64.

Logical Operators

and

X	Y	X and Y
True	False	False
False	True	False
True	True	True
False	False	False

If all variables are True, the outcome is True.

or

X	Y	X or Y
True	False	True
False	True	True
True	True	True
False	False	False

If one variable is True, the outcome is True.

not

X	not X
True	False
False	True

Reverse a variable's logical state.

Logical Operators: Examples



- **and**

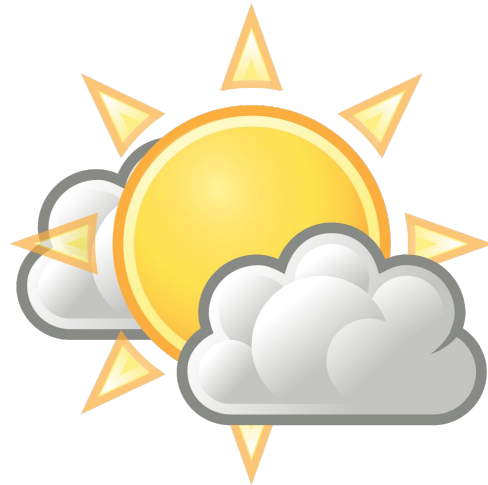
```
if temperature > 0 and temperature < 30:  
    print("The temperature is good")  
else:  
    print("The temperature is bad")
```

- **or**

```
if temperature <= 0 or temperature >= 30:  
    print("The temperature is bad")  
else:  
    print("The temperature is good")
```

- **not**

```
is_sunny = False  
if not is_sunny:  
    print("It is not sunny outside!")  
else:  
    print("It is sunny outside!")
```



A Classic Joke: "The Mathematician's Answer"



Q: Would you like an apple or a banana?

A: Yes

Boolean Expressions



- **Precedence of operations:**

- arithmetic $5 * 7$
- comparison $>=$
- not
- and/or

highest precedence



lowest precedence

- **Example:**

```
25 >= 3 + 2 * 10 and not False
25 >= 23 and not False
True and not False
True and True
True
```

```
# arithmetic
# comparison
# not
# and
```



- The following example can produce an infinite loop.

```
while True:
    body
```

Boolean Variables



You can store expressions that evaluate to True/False into variables.

- The variables `x`, `y`, and `z` are assigned data of type `bool`:

- `x = 1 < 2` # True

- `y = "Michael" == "Michelle"` # False

- `z = True`

- You can use boolean variables and chain them together.

`if x and y:`
body



`if True and False:`
body



`if False:`
body

Section Exercise: “High Low Game”

```
Welcome to the High-Low Game
-----
Round 1
Your number is 8
Do you think your number is higher or lower than the computer's?: lower
You were right! The computer's number was 35
Your score is now 1

Round 2
Your number is 88
Do you think your number is higher or lower than the computer's?: higher
Aww, that's incorrect. The computer's number was 100
Your score is now 1

Round 3
Your number is 63
Do you think your number is higher or lower than the computer's?: higher
You were right! The computer's number was 5
Your score is now 2

Thanks for playing!
```



Game Steps:

- #1)** Randomly generate two numbers from 1 to 100 (inclusive):
 - One number for you, and one number for the computer
- #2)** Game prints your number but not the computer's. You make a guess:
 - Type “**lower**” if you think your number is lower than the computer's
 - Type “**higher**” if you think your number is higher
- #3)** If you guess correctly, you score 1 point!

“High Low Game” Milestones



Milestone #1: Generate two random numbers.

Milestone #2: Get the player's choice of `lower` or `higher`.

Milestone #3: Write the game logic. Compare player's guess vs actual #.

Milestone #4: Play multiple rounds.

Milestone #5: Add a points system.

Extension #1: Safeguard user input.

Extension #2: Conditional ending messages.